

Date:18/09/2019,Wednesday Time:10:00 a.m. to 10:30 a.m. Maximum Marks:20

[illegible]

- Tick the correct answer/s and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

1. If $P(A) = 0.9$, $P(B) = 0.8$ and $P(A \cap B) = 0.75$ then $P(A \cap B') =$ _____.
(i) 0.95 (ii) 0.18 (iii) 0.1 (iv) 0.15
2. If A, B and C are events of a sample space $\mathcal{S}$, then the expression for the events "only A occurs" is _____.
(i) $A \cup B \cup C'$ (ii) $A \cap B' \cap C'$ (iii) $A \cap B \cap C$ (iv) $A \cup B \cap C'$
3. If A and B are two events of sample space $\mathcal{S}$ and $A, B \neq \phi$, then A and B are independent if _____.
(i) $P(A|B) = P(B)$ (ii) $P(B|A) = P(A)$ (iii) $P(A \cup B) = P(A) + P(B)$ (iv) $P(A \cap B) = P(A)P(B)$

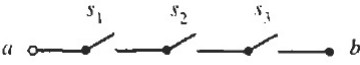
Page 1 of 6

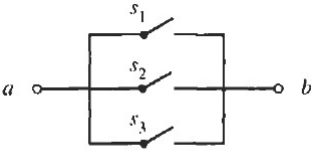
(i) $A_1 \cup A_2 \cup A_3$

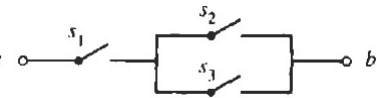
(iii) $A_1 \cap A_2 \cap A_3$

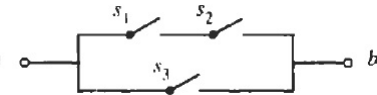
(ii) $(A_1 \cap A_2) \cup A_3$

(iv) $A_1 \cap (A_2 \cup A_3)$

4.  $A_{ab} = \underline{\hspace{2cm}}.$

5.  $A_{ab} = \underline{\hspace{2cm}}.$

6.  $A_{ab} = \underline{\hspace{2cm}}.$

7.  $A_{ab} = \underline{\hspace{2cm}}.$

For questions 8. and 9. consider X is the random variable whose cumulative probability

distribution is given by
$$F(x) = \begin{cases} 0, & x < 0 \\ x + \frac{1}{2}, & 0 \leq x < \frac{1}{2} \\ 1, & x \geq \frac{1}{2} \end{cases}$$

Then,

8. $P\left(X \leq \frac{1}{4}\right) = \underline{\hspace{2cm}}.$

(i) $\frac{1}{2}$

(iii) $\frac{3}{4}$

(ii) $\frac{1}{4}$

(iv) 0

9. $P\left(0 < X \leq \frac{1}{4}\right) = \underline{\hspace{2cm}}.$

(i) $\frac{1}{2}$

(iii) $\frac{3}{4}$

(ii) $\frac{1}{4}$

(iv) 0

For questions 10. and 11. consider X is a continuous random variable with probability

density function
$$f(x) = \begin{cases} kx & 0 < x < 1 \\ 0 & \text{Otherwise,} \end{cases}$$

then

10. $k = \underline{\hspace{2cm}}.$

(i) $\frac{1}{2}$

(iii) 2

(ii) 1

(iv) $\frac{1}{4}$

11. $P(0 < X \leq 2) =$ _____.
- (i) 1 (iii) 0
(ii) $\frac{1}{4}$ (iv) 4
12. If X represents the number of heads in a five tosses of a fair coin, then $P(X = 0) =$ _____.
- (i) $\frac{1}{32}$ (iii) $\frac{5}{32}$
(ii) $\frac{1}{2}$ (iv) $\frac{1}{5}$
13. If $M_X(t)$ is a moment generating function of a random variable X , then $E(X) =$ _____.
- (i) $\left. \frac{d^2}{dt^2} M_X(t) \right|_{t=0}$ (iii) $\frac{d^2}{dt^2} M_X(t)$
(ii) $\left. \frac{d}{dt} M_X(t) \right|_{t=0}$ (iv) $\frac{d}{dt} M_X(t)$
14. If a random variable X follows poisson distribution with parameter λ ($\lambda > 0$) and $P(X = 0) = P(X = 1)$, then $\lambda =$ _____.
- (i) 2 (iii) e
(ii) 0.5 (iv) 1
15. If a random variable X follows binomial distribution with parameters $n = 6$ and $p = 0.4$ then standard deviation of X is _____.
- (i) 6 (iii) 1.2
(ii) 2.4 (iv) 1.44
16. If the probability density function of Beta distributed random variable X is
- $$f(x) = \begin{cases} \frac{1}{12}x^2(1-x), & 0 < x < 1 \\ 0, & \text{Otherwise,} \end{cases}$$
- then $E(X) =$ _____.
- (i) $\frac{3}{5}$ (iii) $\frac{5}{3}$
(ii) $\frac{2}{5}$ (iv) $\frac{5}{2}$

17. If the probability density function of gamma distributed random variable X is

$$f(x) = \begin{cases} \frac{e^{-x} x^{\lambda-1}}{\Gamma(\lambda)} & 0 < x < \infty \\ 0 & \text{Otherwise,} \end{cases} \text{ and } \lambda > 0.$$

then which of the following statement is true? _____.

(i) mean of X < variance of X

(iii) mean of X > variance of X

(ii) mean of X = variance of X

(iv) mean of X = standard deviation of X

18. If a random variable X follows normal probability distribution with parameters $\mu = 0$ and $\sigma^2 = 1$, then $P(X > 1) =$ _____.

(i) 0.5

(iii) 0

(ii) $1 - P(X < -1)$

(iv) $P(X < -1)$

19. For a discrete random variable X , $P(X = x) = \begin{cases} \frac{1}{3}, & x = -1, 0, 1 \\ 0, & \text{Otherwise,} \end{cases}$

Then $E(X) =$ _____.

(i) $\frac{2}{3}$

(iii) 0

(ii) $\frac{1}{3}$

(iv) $\frac{1}{9}$

20. If X is a random variable such that $E(X) = 3, V(X) = 4$, then $P(-2 < X < 8) \geq$ _____.

(i) 0.84

(iii) 0.4

(ii) 0.16

(iv) 0.6

End of MCQ

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Second & Fourth Semester of BSc Physics Examination September 2019
PD156 Statistics-II

Date: 18/09/2019, Wednesday

Time: 10:30 a.m. to 01:00 p.m.

Maximum Marks: 50

Instructions:

- Section I and II must be attempted in TWO ANSWER SHEET.
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.

SECTION I

Q - II Answer the following questions as directed. [20]

1. Define the following terms. [06]

- (i) State the Baye's theorem. (ii) random variable. (iii) axiomatic definition of probability.

2. Attempt any two of the following. questions. [08]

- (i) From a group of 5 women and 7 men, how many different committees consisting of 2 women and 3 men can be formed? What if 2 of the men are feuding and refuse to serve on the committee together?
- (ii) A total of 36 members of a club play tennis, 28 play squash, and 18 play badminton. Furthermore, 22 of the members play both tennis and squash, 12 play both tennis and badminton, 9 play both squash and badminton, and 4 play all three sports. How many members of this club play at least one of three sports?
- (iii) If X represent a number on upper face of a die when it is rolled once, then compute
 (a) $E(X)$ (b) $E(X^2)$ (c) $V(X)$

3. Attempt any one question. [06]

- (i) A company producing electric relays has three manufacturing plants producing 50, 30, and 20 percent, respectively, of its product. Suppose that the probabilities that a relay manufactured by these plants is defective are 0.02, 0.05, and 0.01, respectively.
- (a) If a relay is selected at random from the output of the company, what is the probability that it is defective?
- (b) If a relay selected at random is found to be defective, what is the probability that it was manufactured by plant 2?
- (ii) A binary source generates digits 1 and 0 randomly with probabilities 0.6 and 0.4, respectively. What is the probability that
- (a) two 1's and three 0's will occur in a five-digit sequence?
- (b) at least three 1's will occur in a five-digit sequence?

SECTION II

Q - III Answer the following questions as directed.

[30]

1. (i) Determine the mean and variance of random variable X whose moment generating function is given by $M_X(t) = e^{\lambda(e^t - 1)}$, $\lambda > 0$, where $M_X(t) < \infty$, $|t| \leq T$, for some positive number T .

[03]

- (ii) State Lindeberg-Lévy central limit theorem.

[03]

2. Attempt any three of the following.

[12]

- (i) The number of telephone calls arriving at a switchboard during any 10-minute period is known to be a Poisson random variable X with $\lambda = 2$. Find the probability that during any 10-minute period (a) more than three calls will arrive (b) no calls will arrive.

- (ii) Let a random variable X is the time (in hours) between consecutive traffic accidents

$$\text{with probability density function } f(x) = \begin{cases} \frac{1}{60}e^{-x/60} & x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Find (a) $P(X \leq 60)$ (b) $P(X > 120)$

- (iii) If the probability density function of random variable X is given by

$$f(x) = \begin{cases} \frac{1}{b-a} & a < x < b, \quad a \in R, b \in R \\ 0 & \text{otherwise.} \end{cases}$$

Show that $E(X) = \frac{a+b}{2}$ and $V(X) = \frac{1}{12}(b-a)^2$.

- (iv) A certain machine makes electrical resistors having a mean resistance of 40Ω and a standard deviation of 2Ω . Assuming that the resistance follows a normal distribution and can be measured to any degree of accuracy, what percentage of resistors will have a resistance exceeding 43Ω ? (given $P(Z \leq 1.5) = 0.9332$)

3. Attempt any two of the following.

[12]

- (i) Suppose the proportion of error in code developed by a programmer, which varies from software to software, may be looked upon as a random variable X having the

$$\text{probability density function } f(x) = \begin{cases} \frac{1}{B(2,7)}x(1-x)^6 & 0 < x < 1 \\ 0, & \text{Otherwise} \end{cases}$$

- (a) Find the probability that a software developed by this engineer will contain 30% or more errors. (given $P(X < 0.3) = 0.7447$)

- (b) Find the mean of X .

- (ii) If the probability density function of a random variable X is

$$f(x) = \begin{cases} \frac{1}{2^3\Gamma(3)}x^2e^{-x/2} & x > 0 \\ 0, & \text{Otherwise,} \end{cases}$$

- (a) Find Mean of X (b) Find $P(1.8 < X < 2.4)$

(given the cumulative distribution function $F(2.4) = 0.3374$, $F(1.8) = 0.2275$)

- (iii) If a random variable X has normal distribution with parameters μ and σ . Determine the moment generating function of X .